
What is AWS GuardDuty?

Amazon GuardDuty is a **threat detection** service that continuously monitors your AWS accounts, workloads, and data for **malicious activity** and **unauthorized behavior**.

It helps protect your AWS environment using:

- Machine Learning 
- Threat Intelligence Feeds (AWS, CrowdStrike, Proofpoint)
- Integrated AWS log sources (like VPC Flow Logs, DNS Logs, and CloudTrail)

How Does GuardDuty Work?

It analyzes:

- **VPC Flow Logs:** Network traffic patterns
- **AWS CloudTrail Logs:** User/API activity
- **Route 53 DNS Logs:** DNS queries from instances

Then it applies:

- ML algorithms
 - Known threat signatures
 - Anomaly detection
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Components of GuardDuty

Component	Description
Detector	Core GuardDuty instance that performs monitoring
Findings	Alerts generated when suspicious activity is detected
Threat Lists	Custom IPs/domains to treat as suspicious
Trusted IP Lists	Known safe IPs to ignore
Member Accounts	Centralized monitoring of multiple accounts in Organizations
Suppression Rules	Filters to suppress specific findings

Example Threats Detected

- Crypto mining activity 
 - Unusual API calls
 - Tor-based IP activity 
 - Port scanning
 - Suspicious data exfiltration
 - Unauthorized access to S3 buckets
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Benefits

- **Fully managed** – No infrastructure to manage
- **Real-time detection**

- **Integration with Security Hub**, Lambda, SNS, SIEM tools
 - **Cost-effective** – Charged based on logs analyzed
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